

Engineering Mathematics Study Notes: Boundedness and Convergence Behavior

1. Concept Identification

The notes introduce how sequences behave as they expand indefinitely, categorized into two main properties:

- **Boundedness:** Determining if a sequence is trapped within real boundaries (Upper and Lower bounds).
 - **Convergence Behavior:** Evaluating what happens to the value of the n -th term (u_n) as n approaches infinity (∞). This divides sequences into three types:
 - i. **Convergent:** Headed toward a single, fixed real number.
 - ii. **Divergent:** Exploding toward positive infinity ($+\infty$) or negative infinity ($-\infty$).
 - iii. **Oscillatory:** Bouncing back and forth between different values without settling down.
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2. Mathematical Representations

Boundedness Thresholds

A sequence is **bounded below** if every term is greater than or equal to a lower bound k :

$$u_n \geq k \quad \forall n \in \mathbb{N}$$

A sequence is **bounded above** if every term is less than or equal to an upper bound K :

$$u_n \leq K \quad \forall n \in \mathbb{N}$$

A sequence is strictly bounded if it has both a lower bound and an upper bound:

$$k \leq u_n \leq K$$

Convergence Criterion

A sequence **converges** if its limit as n goes to infinity equals a definitive, single **finite** number L :

$$\lim_{n \rightarrow \infty} u_n = L \quad (\text{where } L \text{ is finite and unique})$$

Divergence Criterion

A sequence **diverges** if its values increase or decrease without limit:

$$\lim_{n \rightarrow \infty} u_n = \pm\infty$$

Limit Inversion Property

When evaluating limits at infinity, any constant divided by an infinitely large number mechanical collapses to zero:

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

3. The Algorithm (Procedural Recipe)

To evaluate the convergence behavior of a fractional rational sequence ($u_n = \frac{P(n)}{Q(n)}$) as shown on the board, apply this algebraic recipe:

Step 1: Identify the Highest Power of n

Look at both the numerator and denominator expressions and find the highest power of the variable n .

Step 2: Factor Out n Forcefully

Pull out that highest power of n from every term in the numerator and denominator to create terms of the form $\frac{\text{constant}}{n}$.

Step 3: Cancel Common Factors

Algebraically cross out the matching n terms outside the parentheses from the top and bottom.

Step 4: Apply the Limit (∞) Conversion

Substitute ∞ into the remaining n variables, converting all fraction components of $\frac{\text{constant}}{\infty}$ explicitly into 0.

Step 5: Classify the System Behavior

- If the result is a clean, single finite number, label it **Convergent**.
 - If the result is $\pm\infty$, label it **Divergent**.
 - If it alternates between values, label it **Oscillatory**.
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4. Operational Mechanics

Mechanical Example 1: Understanding Boundedness

The board displays three basic numeric frameworks to show how bounds function:

Case A: Bounded Below Only

The sequence is given as: 1, 2, 3, 4, 5, ...

The sequence starts at 1 and grows larger forever. It has a floor but no ceiling.

$$u_n \geq 1$$

Therefore, the lower bound is 1.

Case B: Bounded Above Only

The sequence formula is $u_n = -n$, yielding: -1, -2, -3, -4, -5, ...

The sequence starts at -1 and drops smaller forever. It has a ceiling but no floor.

$$u_n \leq -1$$

Therefore, the upper bound is -1.

Case C: Fully Bounded

The sequence is given as: 1, $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, ...

Let's look at the boundaries:

- The first term is 1 (it never gets higher than this).
- As n grows, the fraction gets closer and closer to 0 (it never drops below or touches 0).

$$0 < u_n \leq 1$$

Lower Bound = 0, Upper Bound = 1. The sequence is fully trapped.

Mechanical Example 2: Solving a Convergent Rational Limit

Let's mechanically solve the problem from the board:

$$u_n = \frac{3n + 2}{n + 5}$$

Apply Step 2 by pulling out the highest power of n (which is n^1) from both the top and the bottom expressions:

$$u_n = \frac{n \cdot \left(3 + \frac{2}{n}\right)}{n \cdot \left(1 + \frac{5}{n}\right)}$$

Cancel the matching n terms from the top and bottom:

$$u_n = \frac{3 + \frac{2}{n}}{1 + \frac{5}{n}}$$

Apply the limit process by letting $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} u_n = \frac{3 + \frac{2}{\infty}}{1 + \frac{5}{\infty}}$$

Convert the infinite fractions directly to 0:

$$\lim_{n \rightarrow \infty} u_n = \frac{3 + 0}{1 + 0} = 3$$

Because 3 is a clean, unique finite value, the sequence is **Convergent**.

Mechanical Example 3: Solving a Divergent Limit

Let's mechanically evaluate the second example shown on the board:

$$u_n = n^3$$

Apply the limit directly to the term as n grows infinitely large:

$$\lim_{n \rightarrow \infty} n^3 = (\infty)^3 = \infty$$

Because the value escapes to infinity, the sequence is classified as **Divergent**.

Mechanical Example 4: Identifying Oscillatory Behavior

The lecturer writes down a sequence alternating signs: $1, -1, 1, -1, 1, \dots$

Let's look at the value behavior based on whether the term position (n) is odd or even:

$$u_n = 1 \quad (\text{when } n \text{ is Odd})$$

$$u_n = -1 \quad (\text{when } n \text{ is Even})$$

When you apply a limit to infinity, you cannot predict whether the destination index is odd or even. Because it yields two entirely different values (1 and -1), the limit has a **not unique value**. It is classified as **Oscillatory**.